

# iADAS

## Intelligent Anomaly Detection & Alerting System

AI-Powered Retail Decision Support using Time-Series Forecasting & Generative AI

DATA > FORECAST > DETECT > EXPLAIN > ACT

TEAM ASSIGNMENT: **Team 10**

LIVE DEMO APPLICATION: <https://i-adas.streamlit.app/>

# Detecting "Unusual" Is Not Enough

## Traditional Retail Monitoring

**Reactive** 1M+ Transaction Stream Static Sales Thresholds / Fixed Rules Manual Root-Cause Investigation Delayed Management Action

**Core Bottleneck:** Produces high alert fatigue without explaining why deviations occurred or what action to take.

## iADAS Decision-Support Pipeline

**Proactive** Forecast Expected Demand Baseline User-Configurable Sensitivity Bounds Auto-Isolate Transaction RCA Evidence Gemini Generative Executive Brief

**Core Value:** Reduces the distance between Anomaly -> Understanding -> Executive Action.

“ An alert without context creates another investigation task. iADAS delivers actionable management briefs.

# From Charter to Working AI Product

## REVIEW 1

### Charter & Problem Definition

- Defined retail anomaly problem statement
- Mapped 1M+ transactions scope
- Established core decision workflow

## REVIEW 2

### Data Pipeline & Forecasting

- Cleaned 1,007,913 records
- Built 106-week time-series aggregation
- Trained baseline Prophet model

## REVIEW 3

### Anomaly Engine & GenAI Briefs

- Bound anomaly detection intervals
- Structured Product/Customer RCA
- Integrated Gemini briefing prompt

## REVIEW 4

### Configurable Product & Impact

- Built dynamic sensitivity slider
- Quantified business trade-offs
- Live Streamlit cloud deployment

## FACULTY FEEDBACK INCORPORATED

"Make anomaly sensitivity user-configurable and demonstrate the operational effect on alert volumes."

✓ Fully Implemented

# Why We Designed iADAS This Way



## 1. Forecast Before Detect

Use expected statistical behavior rather than arbitrary static sales numbers.



## 2. Evidence Before AI

Anchor Generative AI strictly on transaction RCA facts to prevent hallucinations.



## 3. Configurable Bounds

Empower management to tune threshold sensitivity to operational capacity.



## 4. Driver Explanation

Isolate top Product, Customer, and Country contributors automatically.



## 5. Human Governance

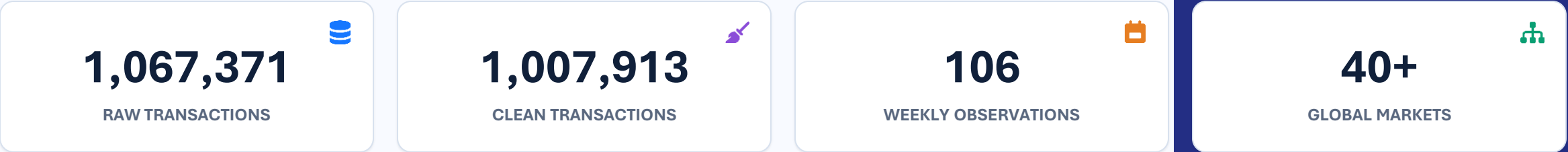
AI provides evidence and briefs; management retains decision authority.



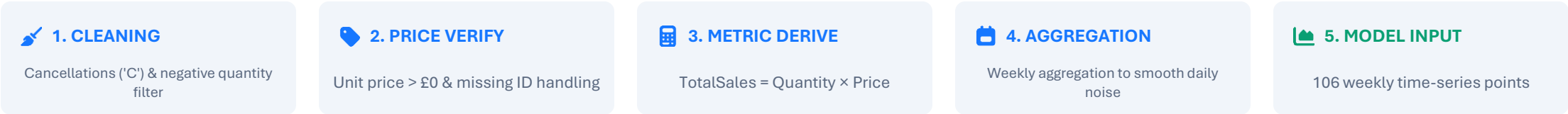
### Core Architectural Design Choice:

"The primary objective of iADAS is not merely model accuracy but reducing executive friction between signal detection and operational execution."

# From 1M+ Transactions to a Forecastable Signal



## Data Refinement & Feature Engineering Pipeline



# Forecasting Defines What "Expected" Looks Like



Weekly Sales Forecast & Prediction Interval Band (£)



AVERAGE WEEKLY SALES



£193,172

MEAN ABSOLUTE ERROR (MAE)



£27,549

~14.2% of Avg Weekly Volume

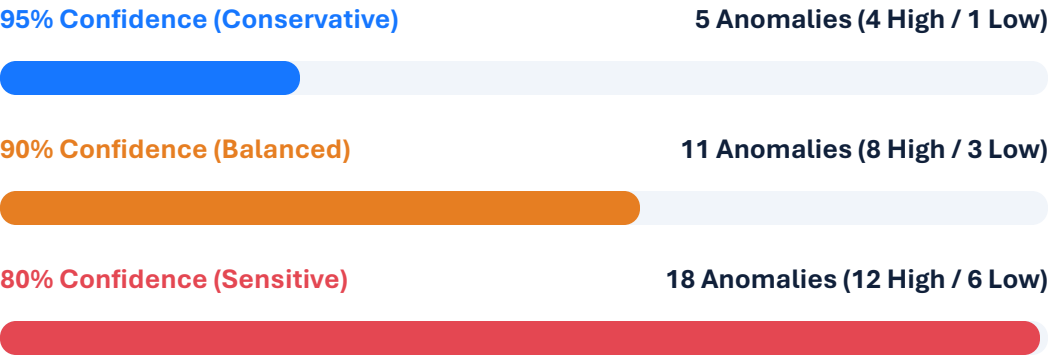
ROOT MEAN SQUARED ERROR



£36,860

# From Fixed Rule to User-Configurable Sensitivity

## Validated Sensitivity vs Anomaly Volume



### Higher Confidence (95%)

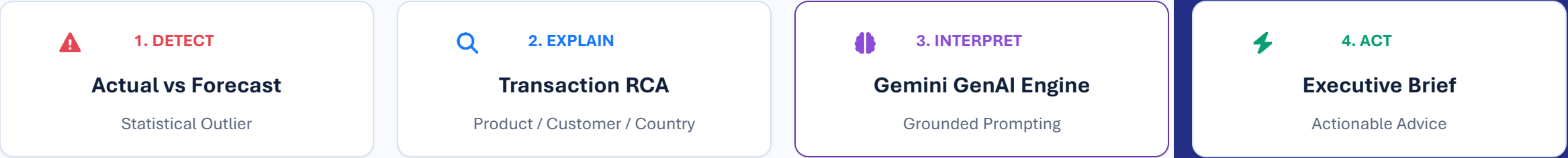
Fewer, highly extreme alerts. Minimizes false alarms and protects investigation bandwidth.

### Lower Confidence (80%)

Catches subtle, early-stage deviations. Increases total anomaly visibility but requires higher audit capacity.

 **Key Takeaway:** Sensitivity is now an explicit business decision parameter rather than a hardcoded rule.

# From Anomaly to Management Explanation



 **Gemini AI Executive Management Brief**

Grounded strictly on extracted RCA data

**EXECUTIVE SUMMARY & DRIVERS**

Sales spike driven predominantly by UK wholesale client (ID 18102) expanding bulk purchases in Regency Cakestand products, accounting for +38% of total weekly volume.

**RECOMMENDED MANAGEMENT ACTION**

Secure inventory allocations for top SKUs to prevent stockout risk; engage account team to lock in recurring quarterly supply contracts.



# From Analytical Pipeline to Working Product



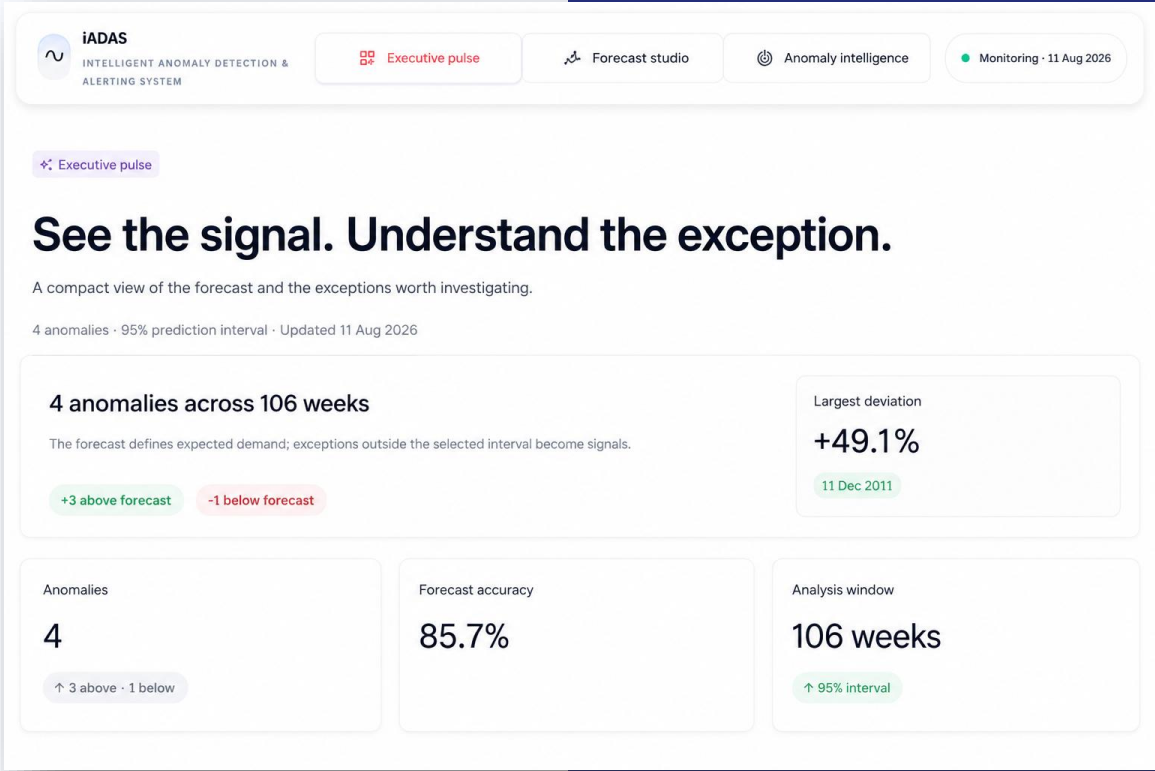
1. Executive Pulse Module

High-level portfolio overview tracking overall sales health and active anomaly counts.
2. Forecast Studio & Sensitivity Slider

Interactive control allowing business users to adjust prediction confidence (95%, 90%, 80%) dynamically.
3. Anomaly Intelligence & GenAI Briefing

Instant transaction RCA drill-down paired with automated Gemini executive briefings.

Launch Live App: <https://i-adas.streamlit.app/>



# What We Delivered - and How Value Will Be Measured

## CHARTER SCOPE DELIVERED

| Module                                | Status                           |
|---------------------------------------|----------------------------------|
| Weekly Time-Series Forecasting        | <div><div>✓</div>Delivered</div> |
| Dynamic Anomaly Detection Engine      | <div><div>✓</div>Delivered</div> |
| Transaction Root-Cause Analysis (RCA) | <div><div>✓</div>Delivered</div> |
| Gemini Executive Brief Generator      | <div><div>✓</div>Delivered</div> |
| Interactive Streamlit Cloud App       | <div><div>✓</div>Delivered</div> |

### 🎯 PROPOSED DEPLOYMENT KPIs

Future Framework

|                            |                                                  |
|----------------------------|--------------------------------------------------|
| Time to Detect (TTD):      | Reduce from weekly reviews to real-time alerts   |
| Time to Investigate (TTI): | Automated RCA saves ~60% analyst triage time     |
| Revenue Exposure Reviewed: | % of total anomaly volume actioned               |
| Action Conversion Rate:    | Briefings resulting in documented manager action |

# From Forecasting to Decision Intelligence



**HUMAN IN LOOP**  
Management retains final decision authority



**GROUNDING AI**  
Strictly uses supplied RCA evidence



**FACT VS HYPOTHESIS**  
Explicitly separates known drivers



**DATA PRIVACY**  
Anonymized customer handles in production



**GOVERNANCE**  
Configurable threshold management



## iADAS Transforms Retail Transactions into Executive Decision Support

By seamlessly uniting time-series forecasting, automated transaction-level root cause analysis, user-configurable sensitivity, and grounded Generative AI briefings, iADAS delivers an end-to-end, responsible AI product.

Team 10

<https://i-adas.streamlit.app/>